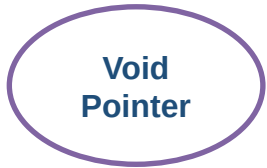


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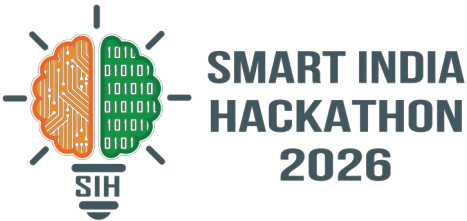
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- **Problem Statement ID – 26068**
- **Problem Statement Title- WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 18**
- **Team Name (Registered on portal)- Void Pointer**



WEATHERGPT: VERIFIED WEATHER INTELLIGENCE

A location-aware, multilingual conversational layer for verified and actionable weather intelligence.



Proposed Solution

- **Intent & Location Agent:** Understands the user's question, language, location, time range, and use case.
- **Multi-Source Retrieval Layer:** Queries relevant weather, forecast, warning, AQI, radar, and historical-data services in parallel.
- **Forecast Verification Agent:** Cross-checks observations and model outputs, resolves source conflicts, tracks freshness, and reports confidence.
- **Evidence-Grounded Response Agent:** Generates concise answers with source, issue time, validity period, location, and uncertainty.
- **Personalized Advisory Layer:** Converts verified data into farmer, commuter, aviation, marine, and disaster-preparedness guidance.
- **Multilingual Voice Interface:** Delivers text, audio, and alert notifications in Indian languages, including low-connectivity use cases.

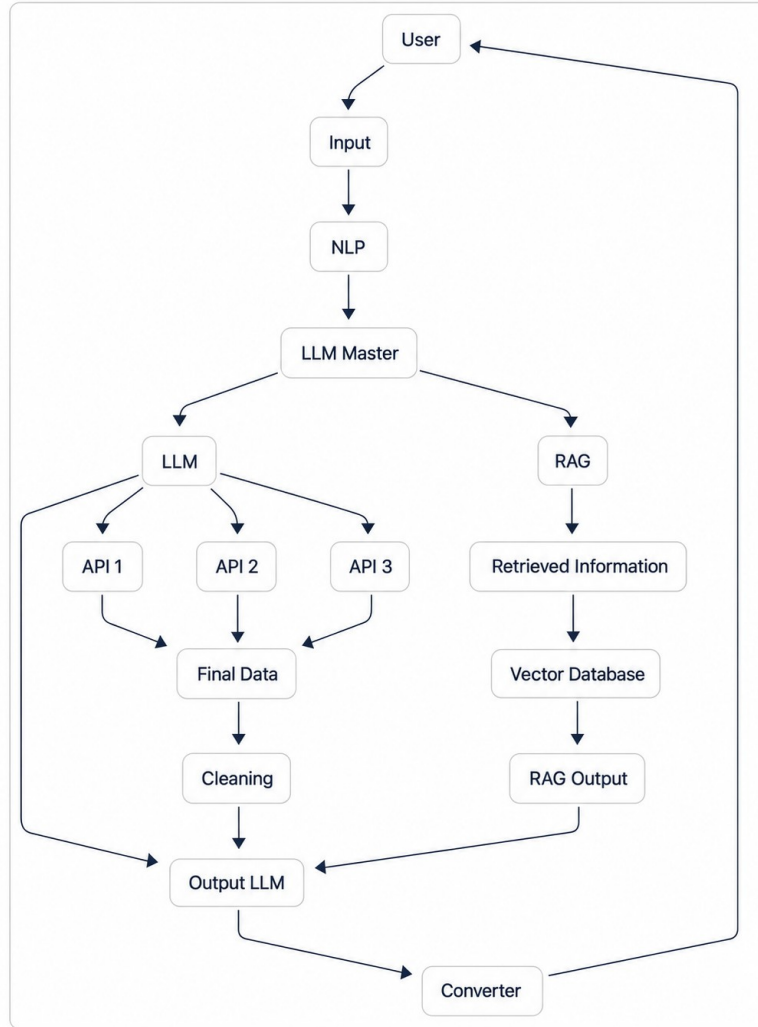
How It Addresses the Problem

- **Unified access:** Consolidates scattered weather portals, bulletins, model outputs, and warning systems into one conversational interface.
- **Local relevance:** Uses the user's precise location and administrative region to provide locally relevant information.
- **Official-first warnings:** Prioritizes IMD warnings and preserves their severity, affected area, issue time, and validity.
- **Historical context:** Uses historical and climate data for anomaly detection, comparisons, and explanations—not as a replacement for numerical forecasts.
- **Inclusive access:** Provides voice and local-language responses for rural and low-literacy users.
- **Trustworthy alerts:** Delivers timely, explainable alerts instead of unverified generic predictions.

Innovation / USP

- **Provenance-aware fusion:** Every answer is linked to its source, timestamp, model, and confidence.
- **Agentic orchestration:** Specialized agents select the right data source and workflow for each query instead of relying on one API.
- **Disagreement detection:** Identifies when providers or models disagree and communicates uncertainty clearly.
- **Location-to-action intelligence:** Converts raw weather values into region-specific recommendations and preparedness actions.
- **One interface, many users:** Supports citizens, farmers, disaster managers, aviation, marine operations, and researchers.
- **Text + voice + alerts:** Makes actionable weather intelligence accessible across languages, literacy levels, and connectivity conditions.

TECHNICAL APPROACH



TECHNOLOGIES USED

CLIENT

React Native, WebSocket

VOICE

Gemini (STT), Groq (TTS)

LANGUAGE

Sarvam AI

LIVE DATA

WeatherAPI, Open-Meteo,
OpenWeatherMap, IMD

RAG

pgvector / PostgreSQL, IMD, ERA5

GENERATION

Ollama (self-hosted)

ALERTS

NDMA Sachet, MQTT, FCM

INFRASTRUCTURE

FastAPI, Redis, Docker, Kubernetes

FEASIBILITY AND VIABILITY

FEASIBILITY

Technical foundations and implementation readiness

- Uses existing AI/LLM services, weather APIs, satellite feeds, GPS, and mobile technologies; no unproven hardware dependency.
- Forecast differences and data errors are managed through verified sources, model comparison, freshness checks, and forecast-confidence display.
- Provider-agnostic backend adapters allow weather-data providers to be added or replaced without rebuilding the complete application.
- Recommendations become more useful when the system understands the user's location, activity, preferences, and previous queries.

VIABILITY

Trust, adoption and a clear path to scale

- User privacy is protected through secure login, encryption, and limited collection of location and personal data.
- Future versions can notify users about significant weather changes or risks instead of waiting for users to check the application.
- The system can grow from a local service into a regional or national platform by adding languages, locations, and industry-specific features.
- Users receive one relevant answer instead of manually interpreting multiple weather parameters and charts.

IMPACT AND BENEFITS

POTENTIAL IMPACT

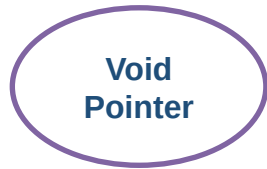
Who benefits—and why it matters

- WeatherGPT delivers clear, location-based weather intelligence that helps citizens make safer decisions while connecting authorities, researchers, and industries through one common platform.
- Rural and low-literacy users can access weather information through voice-based interaction.
- By combining accessibility, personalisation, and reliable data, WeatherGPT has strong potential for large-scale social and economic impact.

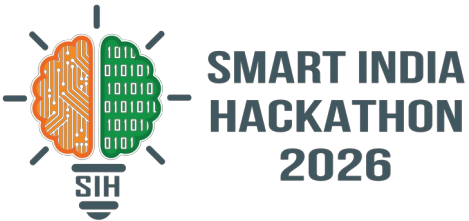
BENEFITS OF THE SOLUTION

Safety, resilience and sustainable action

- It improves public safety by providing timely warnings and practical guidance during extreme weather events.
- WeatherGPT transforms reliable weather and climate data into timely actions that protect lives, reduce losses, support sustainable development, and strengthen community resilience.



RESEARCH AND REFERENCES



Selected official portals and technical references for WeatherGPT.



OFFICIAL SOURCES

Warnings • observations • standards

[IMD](#) • [WMO](#) • [NOAA](#) • [NASA](#)



WEATHER APIs

Conditions • forecasts • alerts

[OpenWeather](#) • [WeatherAPI](#)



AI & DEVELOPMENT

LLMs • RAG • agent workflows

[OpenAI](#) • [Hugging Face](#) • [LangChain](#) • [LangGraph](#)



RESEARCH PAPERS

Verification • conversational AI • human factors

[Google Scholar](#) • [IEEE Xplore](#)



CLIMATE DATA

Baselines • anomalies • vulnerability

[Copernicus](#) • [World Bank Climate Data](#)

Together, these references support reliable, explainable and locally relevant weather answers.



THANK YOU

QUESTIONS?

Team Void Pointer • Team ID 18

PS 26068 — WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information

Theme: Disaster Management • Category: Software • We welcome your questions and feedback.